Terrain classification

Real BorealTC data | One-page project summary

Problem and dataset

Classify the terrain beneath a mobile robot from short sensor time series. Inspired by Choi and Thomasson's tactile-learning study [1], this project compares handcrafted statistics with learned temporal features to support terrain-aware robot navigation. Real BorealTC measurements [2,3] provide 71 single-terrain recordings: asphalt, flooring, ice, loam and snow. Ten channels combine angular velocity (3), acceleration (3), motor current (2) and wheel speed (2). The supplied IMU grid is 100 Hz. We use 4,155 non-overlapping 170-sample windows (1.7 s), split into 2,654 training, 571 validation and 930 test windows.

The supplied PDFs contain no raw SAIL-R data, and a public download was not located. BorealTC was selected as a real, openly available alternative. It uses a wheeled Husky robot and proprioceptive sensors, not the paper's six tactile taxels; this is an adaptation, not an exact reproduction. The folder label SANDY_LOAM is preserved although the authors describe silty loam.

Approach and implementation

Recordings are assigned to train/validation/test within each terrain before windowing (seed 42). Wheel channels are aligned to IMU timestamps using the latest preceding observation within common time coverage. Normalization is fitted on training data only. Six statistics per channel (mean, standard deviation, minimum, maximum, RMS and first-difference RMS) yield 60 features.

Models: balanced RBF SVM on 60 statistics; four-block 1D CNN; and CNN + statistics. CNN channels: 10 input, 24 per block, 240 flattened, 64 hidden, 5 outputs. Adam, weighted cross-entropy and validation early stopping train the networks. SVM hyperparameters and the final model are selected by validation macro-F1.

Model	Val. macro-F1	Test accuracy	Test macro-F1
Majority baseline	0.042	23.8%	0.077
RBF SVM	0.688	84.1%	0.817
1D CNN	0.690	79.6%	0.770
1D CNN + features	0.726	86.3%	0.845

Conclusions and limitations

The validation-selected model is **cnn + features**; it achieves **86.3% test accuracy** and **0.845 test macro-F1** on unseen recordings. These are measurements from this project, not quoted paper results. The implementation demonstrates a reproducible real-data pipeline; conclusions are restricted to the selected BorealTC split.

Only one seed/split was evaluated; asphalt has five validation windows. Performance does not establish tactile-sensor accuracy or cross-robot generalization. Saved models, an offline real-data demo, setup instructions and machine-readable results accompany the code.

References

[1] H. Choi and R. Thomasson. Terrain Classification for Small Legged Robots Using Deep Learning on Tactile Data. Supplied Stanford report, 2020. cs229.stanford.edu/proj2020spr/report/Choi_Thomasson.pdf

[2] D. LaRocque et al. Proprioception Is All You Need: Terrain Classification for Boreal Forests. IROS, 2024, pp. 11686-11693. doi:10.1109/IROS58592.2024.10801407.

[3] BorealTC dataset: github.com/norlab-ulaval/BorealTC. Pinned revision 0146dcd9fa08c34a9075a80448ac04c0a947b568. Dataset README declares CC0 1.0. Retrieved 8 October 2026.